

EMISSION CALCULATION METHODOLOGY

Estimating Carbon

April 27 2026

OVERVIEW

1. What we measure

Carbon emissions calculated in gCO₂e; a standard measurement that converts different greenhouse gases into a metric representing their global warming potential relative to 1 gram of CO₂

2. Methodology of Calculation

Calculations align with the Green Software Foundation's methodology for Software Carbon Intensity (SCI):

$$SCI = ((E \times I) + M) \div R$$

Our methodology is below:

- Measure amount of processed input and output per prompt in tokens (R)
- Convert tokens into model-specific energy cost in kwh (E)
- Apply local grid carbon intensity factor to energy cost to produce carbon impact in gCO₂e (I)
- Currently manufacturing emissions are not considered in our calculations (M)

3. Accuracy of data

Our carbon emission calculator is based on the best data we could find, but AI model providers do not publish transparent per-token energy or CO₂ data. Most available values come research estimates. Values here should be treated as indicative, not exact. The emission data we provide is purely from generations prompted by the user (inference). The carbon footprint of training specific models is not taken into consideration at this time. In the case where specific models are not fully identified, a close average model is used to best estimate the emissions

A major limitation faced with the SCI calculations was the lack of certainty of which AI data centre was processing our request. The Grid Carbon Intensity of a nation varies hugely; Sweden has a low 35 gCO₂e / kwh whilst South Africa has a massive 713 gCO₂e / kwh. As such, the global average of 471gCO₂e/kwh was used.

4. Data sources

The majority of our data comes from an interactive tool that showcases data from the paper

"How Hungry is AI? Benchmarking Energy, Water, and Carbon Footprint of LLM Inference" by Jegham et al. 2025

The link is provided:

<https://app.powerbi.com/view?r=eyJrIjojZjVmOTI0MmMtY2U2Mi00ZTE2LTk2MGYtY2ZjNDMzODZkMjlmIiwidCI6IjQyNmQyYThkLTljY2QtNDI1NS04OTNkLTA2ODZhMzJjMTY4ZCIsImMiOiF9>

The models with tokens that are from this website are as follows:

- GPT o1
- GPT o3
- GPT4o
- GPT4o Mini
- GPT 4
- GPT 4.1
- GPT 4 Turbo
- GPT4.5
- GPT5
- GPT5 nano
- GPT5 mini
- Claude Haiku 4.5
- Claude Sonnet 4
- Claude Sonnet 4.5
- Claude Sonnet 4.6
- Claude Opus 4.5
- Claude Opus 4.6
- Claude Opus 4.5
- Gemini 2.5
- Gemini 2.5 Pro

In cases where the model was available in copilot but not available in the tool, the data was sourced from the Climate Aligned website: <https://www.climatealigned.co/tools/ai-footprint-calculator>

The models taken from that site are below

- Gemini 3.1 Pro
- Gemini 3 Flash

5. Usage of data and conversion to emissions

The data we obtained came in two respects: medium prompts, and long prompts.

A medium prompt gave 1000 tokens of input and produced 1000 tokens of output whilst a long prompt gave 10,000 tokens of input and produced 1500 tokens of output. This data gave rise to a tiered approach to our conversions, allowing us to scale the emissions based on the input and output token count.

However, in some cases giving a large context window actually reduced emissions compared to a smaller context window. This meant that any gradient based approach was invalid – a negative gradient would eventually tend to 0 gCO₂e as the token number increased. As such, we took the intersection of the two data fields as a solid cut; any tokens before the 2000 mark are processed with the respective emission, and anything above that is taken with the 11500 emission. This would, in the worst case, produce double

the emissions that actually occurred. This was the best approach we found, realizing that in the context of estimating emissions produced by AI models, variations of 10x were not uncommon between websites claiming to report on the same model. By using a reputable and academically researched source, ensuring any error was always to the positive, and disclaiming our estimations as that – estimations – we believe this is the best approach to this current issue. We are currently in conversation with the researcher responsible for the paper that we took the data from and hope to be able to access their full dataset soon to improve this linear approach. Any contact with the researcher will be provided to UsTwo and Paolo to ensure cohesiveness for future development.

6. Further contact:

For any further contact regarding emissions, methodology or conversion, please contact:

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